Project pitch

Autonomous Terrain Navigation Using Sentinel-2 Satellite Imagery with Local Sensing and Dynamic Replanning

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Objectives

Comparative Benchmarking

Design controlled experiments to evaluate the local-sensing approach against global Dijkstra and A* baselines, measuring path cost ratios and computational efficiency.

Robustness & Noise Analysis

Stress-test the navigation system by introducing artificial Gaussian noise into the terrain data, simulating sensor inaccuracies and atmospheric artifacts.

Dynamic Agent Navigation

Implement a local-sensing robotic agent with dynamic replanning capabilities, enabling path discovery under partial observability and real-time cost conflict resolution.

Multi-Biome Terrain Modeling

Construct realistic terrain cost maps from Sentinel-2 Level-2A imagery across geographically diverse locations, utilizing NDVI and NDWI indices for granular classification.

Timeline

Week 1

Developed a Python pipeline to transform Sentinel-2 satellite data into validated terrain cost maps using NDVI and NDWI analysis.

Week 2

Developed an 8-connected grid navigation system using A* and Dijkstra algorithms, featuring an autonomous agent with local sensor-based map discovery.

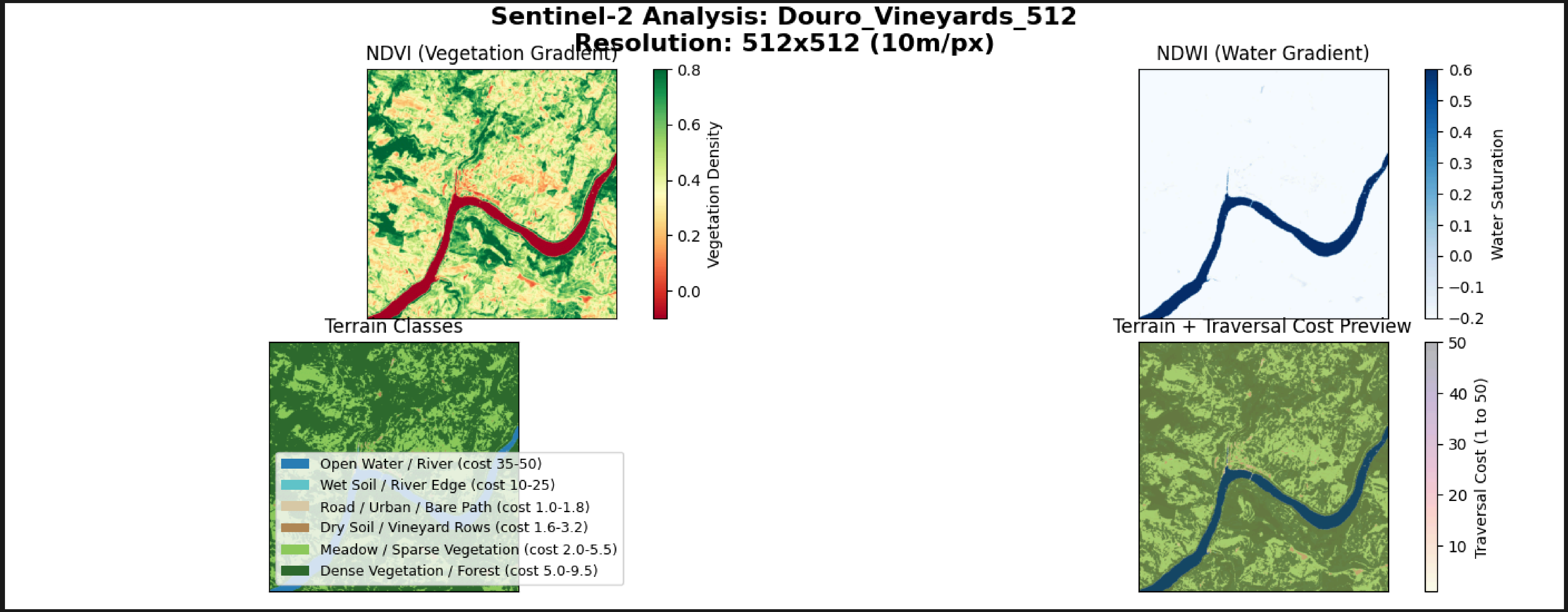
Week 3

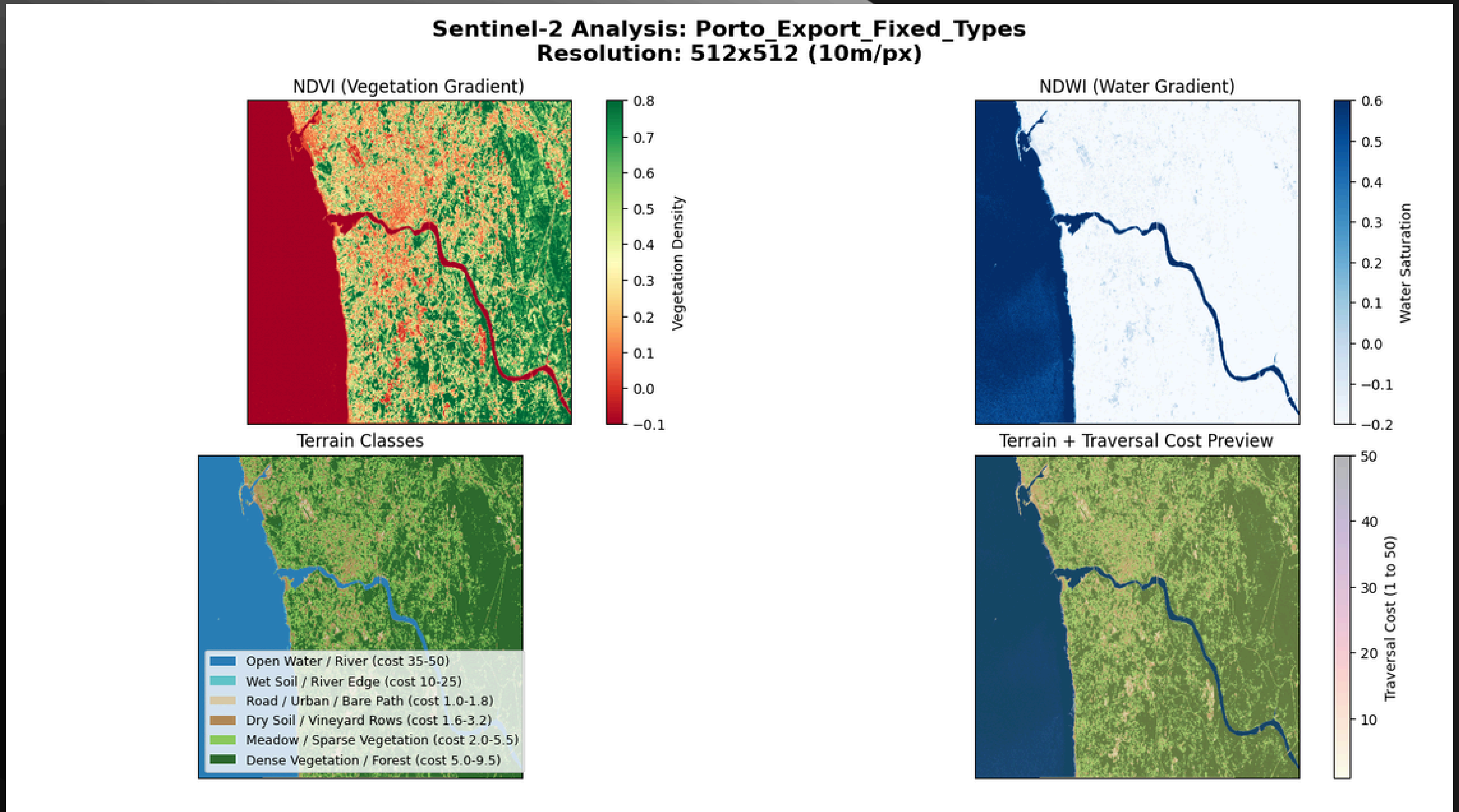
Integrated dynamic A* replanning with a Pygame visualization to simulate real-time robot navigation and sensor-based map discovery on Sentinel-2 data.

Week 4

Conducted large-scale testing across eight scenes, comparing global versus local sensing and analyzing the impact of sensor radius and sensor noise on pathfinding performance.

Cost maps

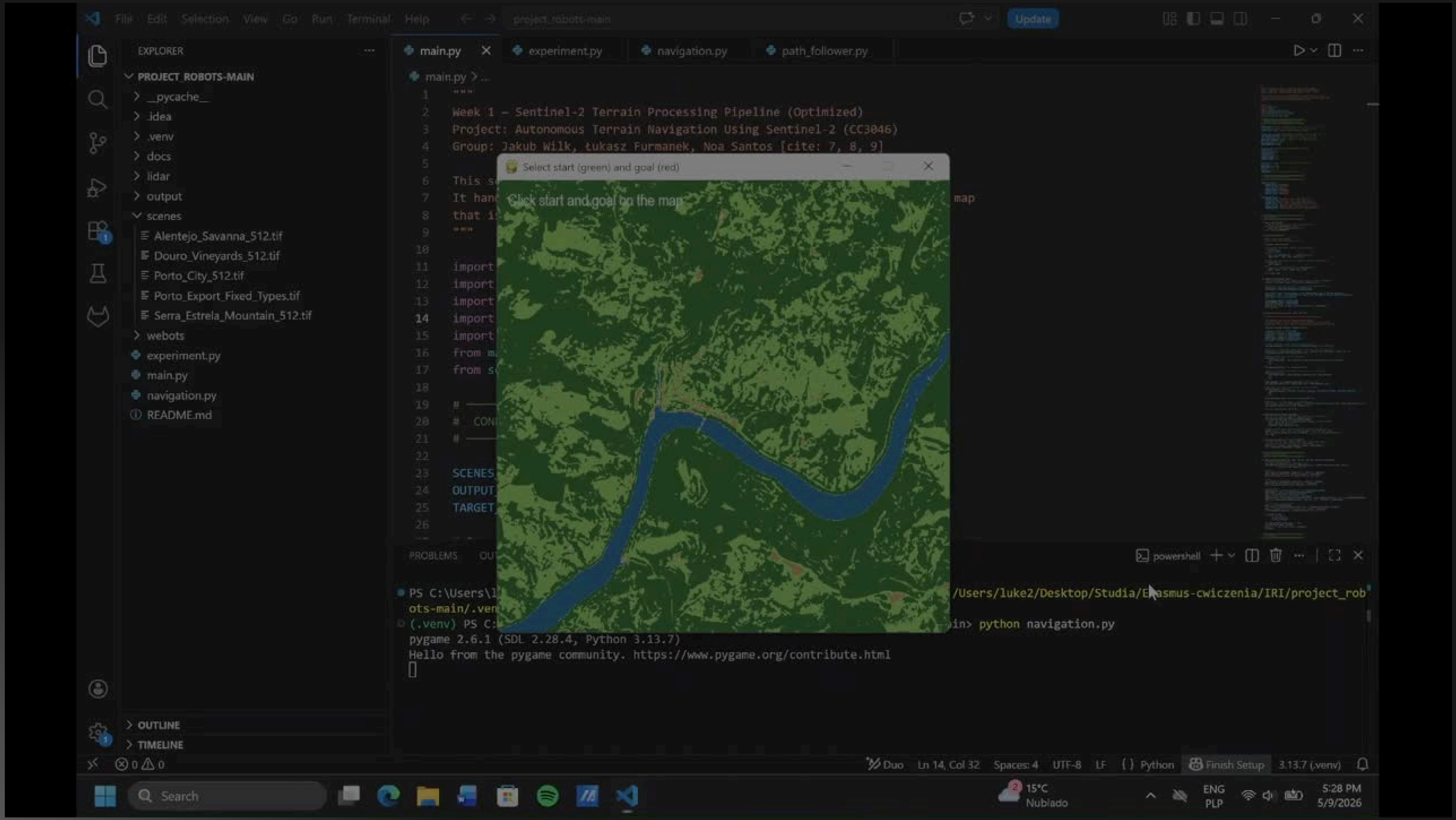




Cost map of Porto and of Douro Valley

- Data Source: Transforms multispectral Sentinel-2 bands into 512x512 navigational grids.
- Index-Based Logic: Uses NDVI and NDWI to classify terrain (Forest, Meadow, Road, Water, etc.).
- Cost Scaling: Assigns traversal costs from 1.0 (Roads) to 50.0 (Open Water) based on vegetation density and moisture.
- Purpose: Provides a validated, continuous cost environment for A* and Reinforcement Learning pathfinding.

Vizualization



Thank you for your attention!

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